DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LABORATORY WORKBOOK

PROGRAM – B. Tech (AI&ML)

23AI&ML6351: Big Data Analytics Lab

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VELAGAPUDIRAMAKRISHNA SIDDHARTHA ENGINEERING COLLEGE

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BIG DATA ANALYTICS LAB WORKBOOK

STUDENT NAME	
REG. NO	
YEAR	
SEMESTER	
FACULTY	

No. of. Experiments Recorded: …….

Marks Awarded: …….

Signature of the Faculty **Signature of the Head of the Dept.**

Course Outcomes

20AI&ML6351: BIG DATA ANALYTICS LAB
Mapping of Course Outcomes with Program Outcomes:

Course Code	Course Name	Course outcomes	Program Outcomes
23AI&ML6351	**BIG DATA ANALYTICS LAB**	1. Apply the concepts of Big Data Analytics and Hadoop components used for managing big data. 2. Apply Hadoop MapReduce programs to perform distributed data processing. 3. Apply Pig and Hive tools to process, manage, and analyze structured and unstructured data in Hadoop. 4.Apply Spark and Sqoop concepts to handle and analyze large-scale data in Hadoop.	PO1, PO2, PO3, PSO1, PSO2

Mapping of individual CO with corresponding PO

	Program Outcomes (1- Low, 2-Medium, 3-High)														
Course Code: Course Name	**Course Outcomes**	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
23AI&ML6351: Big Data Analytics Lab	**CO1**	3	2	1										3	2
	CO2	1		2		3								3	3
	CO3		1		2	3								3	3
	CO4		1		2	3								3	3

COURSE CONTENT

S.No	Experiments	Page No.
Task 1	Installing Hadoop Environment and executing basic shell commands	
Task 2	Program to implement File management operations in Hadoop.	
Task 3	Program to implement Data Import and Export operations between HDFS and Local File System	
Task 4	Hadoop Map Reduce program to find the number of times each word is repeated in a given text file	
Task 5	Program to implement Weather analysis of hot and cold temperatures.	
Task 6	Program to implement a Map Reduce program to identify “common friends” among all pairs of users.	
Task 7	Install Pig and write Pig Latin scripts to load, store and filter data.	
Task 8	Apache Spark program to perform transformation for year-wise maximum temperature analysis.	
Task 9	Develop a MapReduce program to analyse Titanic ship data and to find the average age of the people (both male and female) who died in the tragedy. How many persons survived in each class?	
Task10	Write a program to implement User Defined Functions for filtering unwanted data in Pig?	

ACADEMIC YEAR 2025-2026

23AI&ML6351

BIG DATA ANALYTICS LAB

CONTINUOUS EVALUATION REPORT

S No	Date	Experiment Name	Pre-Lab (2M)	In-Lab				Post-Lab (2M)	Viva Voice (1M)	Total (10M)	Faculty signature
				LOGIC (2M)	EXECUTION (1M)	RESULT (1M)	ANALYSIS (1M)				
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											

Task 1: Hadoop Installation and shell commands.

Date of the Session: ----/-----/------ Time of the Session: to

Pre Lab:

1. What is distributed Systems?

2. What are the prerequisites of the Hadoop environment?

3. What is Hadoop?

4. What do you mean by Meta-data in Hadoop?

5. What is the Difference between Traditional RDBMS and Hadoop?

In Lab:
Installing Hadoop Environment and executing basic shell commands

Task 1 : Installing Hadoop environment

1.Install VirtualBox 7.0 on the desktop

https://www.oracle.com/in/virtualization/technologies/vm/downloads/virtualbox downloads.html

2. Install cloudera and load it into the virtual box and customize the settings

https://downloads.cloudera.com/demo_vm/virtualbox/cloudera-quickstart-vm-5.13.0-0-virtualbox.zip

3. Click start

4. A new environment of cloudera that includes a hadoop environment will be displayed. Executing

basic shell commands

Task 2: Execute basic shell commands

a. To check the present working directory.
b. To list the files in the directory .
c. To create a new directory.
d. To change the directory to the newly created directory.
e. To create a new file.
f. To see the contents of the file.
g. To remove a directory.

1. Type pwd –pwd is the command to check the present working directory.

```
[cloudera@quickstart ~]$ pwd
/home/cloudera
```

2. Now check the available list of files in that directory using the ls command.

```
[cloudera@quickstart ~]$ ls
cloudera-manager  Documents  enterprise-deployment.json  lib      Pictures   Videos
cm_api.py         Downloads  express-deployment.json     Music    Public     workspace
Desktop           eclipse    kerberos                    parcels  Templates
```

3.Create a new folder /directory using the mkdir command followed by the directory name. (give the name of the directory as your name)

```
[cloudera@quickstart ~]$ mkdir exercise1
[cloudera@quickstart ~]$ ls
cloudera-manager  Documents  enterprise-deployment.json  kerberos  parcels   Templates
cm_api.py         Downloads  exercise1                   lib       Pictures  Videos
Desktop           eclipse    express-deployment.json     Music     Public    workspace
```

```
[cloudera@quickstart ~]$ mkdir navyasri
[cloudera@quickstart ~]$ ls
cloudera-manager  Documents  enterprise-deployment.json  kerberos  navyasri  Public     workspace
cm_api.py         Downloads  exercise1                   lib       parcels   Templates
Desktop           eclipse    express-deployment.json     Music     Pictures  Videos
```

4. Now change the directory to the newly created directory using the cd command

```
[cloudera@quickstart ~]$ cd navyasri
[cloudera@quickstart navyasri]$
```

5. Create a new file named friends.txt using gedit command

gedit family.txt

```
[cloudera@quickstart navyasri]$ gedit family.txt
```

6. To see the contents present in the file use the cat command

cat family.txt

```
[cloudera@quickstart navyasri]$ cat family.txt
Santhosh
Jhansi Priya
NavyaSri
Abhinav
```

7. To come back to the previous directory use cd ..

8. To remove the directory use rmdir command followed by the directory name rmdir

exercise1

```
[cloudera@quickstart navyasri]$ cd ..
[cloudera@quickstart ~]$ rmdir exercise1
[cloudera@quickstart ~]$ ls
cloudera-manager  Documents  enterprise-deployment.json  lib       parcels   Templates
cm_api.py         Downloads  express-deployment.json     Music     Pictures  Videos
Desktop           eclipse    kerberos                    navyasri  Public    workspace
[cloudera@quickstart ~]$
```

Post Lab:

1. What are the three modes in which Hadoop can run?

2. What are the different Hadoop configuration files?

3. What are the core components of Hadoop?

4. What are the advantages of Distributed Computing?

5. What is Throughput and how does HDFC provides good Throughput?

(For Evaluator's use only)	
Comment of the Evaluator (if Any)	Evaluator's Observation Marks Secured:_____ out of _____ Signature of the Evaluator

Task 2: File management operations in Hadoop.

Date of the Session: ----/-----/------ Time of the Session: to

Pre Lab:

1. What are the differences between regular File System and HDFS?

2. Which command will help you find the status of blocks and File System health?

3. Difference between parallel and distributed computing?

4. Which commands are used to copy file from one directory to another directory?

5. Explain Main features of HDFS?

In-Lab:

Task 1 : To implement directory commands

a. Create a directory in hdfs using the command.
b. List all files and directories in the hdfs root.
c. List recursively all files and directories in the entire HDFS file system
d. List recursively a specific path.
e. Remove a file or directory
f. Copy files within HDFS
g. Rename files in HDFS
h. Check disk usage in HDFS
i. Directory Commands

Write the commands one below the other to implement over the directory for execution

1. Create a directory in hdfs using the command

hdfs dfs -mkdir -p /user/cloudera/mydir : Creates a new folder mydir in HDFS under /user/cloudera.

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir -p /user/cloudera/mydir
```

2. List all files and directories in the hdfs root.

hdfs dfs –ls

```
[cloudera@quickstart ~]$ hdfs dfs -ls
Found 1 items
drwxr-xr-x   - cloudera cloudera          0 2026-01-19 22:26 mydir
```

3. List recursively all files and directories in the entire HDFS file system

```
[cloudera@quickstart ~]$ hdfs dfs -ls -R /
```

```
-rwxrwxrwx   1 oozie    supergroup   26028391 2017-10-23 09:17 /user/oozie/share/lib/pig/pig-0.12.0-cdh5.13.0.jar
-rwxrwxrwx   1 oozie    supergroup    9429261 2017-10-23 09:17 /user/oozie/share/lib/pig/pig.jar
-rwxrwxrwx   1 oozie    supergroup     386264 2017-10-23 09:17 /user/oozie/share/lib/pig/piggybank.jar
-rwxrwxrwx   1 oozie    supergroup     995968 2017-10-23 09:17 /user/oozie/share/lib/pig/snappy-java-1.0.4.1.jar
drwxrwxrwx   - oozie    supergroup          0 2017-10-23 09:17 /user/oozie/share/lib/sqoop
-rwxrwxrwx   1 oozie    supergroup     224277 2017-10-23 09:17 /user/oozie/share/lib/sqoop/ant-contrib-1.0b3.jar
-rwxrwxrwx   1 oozie    supergroup      36455 2017-10-23 09:17 /user/oozie/share/lib/sqoop/ant-eclipse-1.0-jvm1.2.jar
-rwxrwxrwx   1 oozie    supergroup     181321 2017-10-23 09:17 /user/oozie/share/lib/sqoop/avro-mapred-hadoop2.jar
-rwxrwxrwx   1 oozie    supergroup     477006 2017-10-23 09:17 /user/oozie/share/lib/sqoop/avro.jar
-rwxrwxrwx   1 oozie    supergroup      58160 2017-10-23 09:17 /user/oozie/share/lib/sqoop/commons-codec-1.4.jar
-rwxrwxrwx   1 oozie    supergroup     241367 2017-10-23 09:17 /user/oozie/share/lib/sqoop/commons-compress-1.4.1.jar
-rwxrwxrwx   1 oozie    supergroup     109043 2017-10-23 09:17 /user/oozie/share/lib/sqoop/commons-io-1.4.jar
-rwxrwxrwx   1 oozie    supergroup     267634 2017-10-23 09:17 /user/oozie/share/lib/sqoop/commons-jexl-2.1.1.jar
-rwxrwxrwx   1 oozie    supergroup     434678 2017-10-23 09:17 /user/oozie/share/lib/sqoop/commons-lang3-3.4.jar
-rwxrwxrwx   1 oozie    supergroup      62050 2017-10-23 09:17 /user/oozie/share/lib/sqoop/commons-logging-1.1.3.jar
-rwxrwxrwx   1 oozie    supergroup   14582557 2017-10-23 09:17 /user/oozie/share/lib/sqoop/fastutil-6.3.jar
-rwxrwxrwx   1 oozie    supergroup     706710 2017-10-23 09:17 /user/oozie/share/lib/sqoop/hsqldb-1.8.0.10.jar
-rwxrwxrwx   1 oozie    supergroup      36519 2017-10-23 09:17 /user/oozie/share/lib/sqoop/jackson-annotations-2.3.1.
jar
-rwxrwxrwx   1 oozie    supergroup     197986 2017-10-23 09:17 /user/oozie/share/lib/sqoop/jackson-core-2.3.1.jar
-rwxrwxrwx   1 oozie    supergroup     227500 2017-10-23 09:17 /user/oozie/share/lib/sqoop/jackson-core-asl-1.8.8.jar
-rwxrwxrwx   1 oozie    supergroup     914311 2017-10-23 09:17 /user/oozie/share/lib/sqoop/jackson-databind-2.3.1.jar
-rwxrwxrwx   1 oozie    supergroup     668564 2017-10-23 09:17 /user/oozie/share/lib/sqoop/jackson-mapper-asl-1.8.8.j
ar
-rwxrwxrwx   1 oozie    supergroup    2201101 2017-10-23 09:17 /user/oozie/share/lib/sqoop/kite-data-core.jar
-rwxrwxrwx   1 oozie    supergroup    1807934 2017-10-23 09:17 /user/oozie/share/lib/sqoop/kite-data-hive.jar
-rwxrwxrwx   1 oozie    supergroup    1773731 2017-10-23 09:17 /user/oozie/share/lib/sqoop/kite-data-mapreduce.jar
-rwxrwxrwx   1 oozie    supergroup    1772114 2017-10-23 09:17 /user/oozie/share/lib/sqoop/kite-hadoop-compatibility.
jar
-rwxrwxrwx   1 oozie    supergroup      19827 2017-10-23 09:17 /user/oozie/share/lib/sqoop/opencsv-2.3.jar
-rwxrwxrwx   1 oozie    supergroup      29555 2017-10-23 09:17 /user/oozie/share/lib/sqoop/paranamer-2.3.jar
-rwxrwxrwx   1 oozie    supergroup     106447 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-avro.jar
-rwxrwxrwx   1 oozie    supergroup     952336 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-column.jar
-rwxrwxrwx   1 oozie    supergroup      41084 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-common.jar
-rwxrwxrwx   1 oozie    supergroup     278925 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-encoding.jar
-rwxrwxrwx   1 oozie    supergroup     384617 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-format.jar
-rwxrwxrwx   1 oozie    supergroup     212438 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-hadoop.jar
-rwxrwxrwx   1 oozie    supergroup     927865 2017-10-23 09:17 /user/oozie/share/lib/sqoop/parquet-jackson.jar
-rwxrwxrwx   1 oozie    supergroup      26084 2017-10-23 09:17 /user/oozie/share/lib/sqoop/slf4j-api-1.7.5.jar
-rwxrwxrwx   1 oozie    supergroup     995968 2017-10-23 09:17 /user/oozie/share/lib/sqoop/snappy-java-1.0.4.1.jar
-rwxrwxrwx   1 oozie    supergroup    1060870 2017-10-23 09:17 /user/oozie/share/lib/sqoop/sqoop-1.4.6-cdh5.13.0.jar
-rwxrwxrwx   1 oozie    supergroup     717179 2017-10-23 09:17 /user/oozie/share/lib/sqoop/sqoop-test-1.4.6-cdh5.13.0
.jar
-rwxrwxrwx   1 oozie    supergroup     717179 2017-10-23 09:17 /user/oozie/share/lib/sqoop/sqoop-test.jar
-rwxrwxrwx   1 oozie    supergroup    1060870 2017-10-23 09:17 /user/oozie/share/lib/sqoop/sqoop.jar
-rwxrwxrwx   1 oozie    supergroup      94672 2017-10-23 09:17 /user/oozie/share/lib/sqoop/xz-1.0.jar
drwxrwxrwx   - root     supergroup          0 2017-10-23 09:16 /user/root
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user/spark
drwxrwxrwx   - spark    supergroup          0 2026-01-19 22:29 /user/spark/applicationHistory
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var/lib
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var/lib/hadoop-hdfs
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var/lib/hadoop-hdfs/cache
drwxr-xr-x   - mapred   supergroup          0 2017-10-23 09:17 /var/lib/hadoop-hdfs/cache/mapred
drwxr-xr-x   - mapred   supergroup          0 2017-10-23 09:17 /var/lib/hadoop-hdfs/cache/mapred/mapred
drwxrwxrwt   - mapred   supergroup          0 2017-10-23 09:17 /var/lib/hadoop-hdfs/cache/mapred/mapred/staging
drwxrwxr-t   - yarn     mapred              0 2017-10-23 09:15 /var/log
drwxr-xr-x   - hdfs     mapred              0 2017-10-23 09:15 /var/log/hadoop-yarn
drwxrwxrwt   - yarn     mapred              0 2017-10-23 09:15 /var/log/hadoop-yarn/apps
```

4. List recursively a specific path.

```
[cloudera@quickstart ~]$ hdfs dfs -ls -R /user/cloudera/mydir
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 22:53 /user/cloudera/mydir/file.txt
```

5. Remove a file or directory

To remove file use the –rm command

Syntax: hdfsdfs –rm [path]

```
[cloudera@quickstart ~]$ hdfs dfs -rm /user/cloudera/mydir/file.txt
Deleted /user/cloudera/mydir/file.txt
```

After deleting check the list of files

```
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/mydir
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 22:59 /user/cloudera/mydir/file1.txt
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 23:03 /user/cloudera/mydir/friends.txt
```

6. Copy files within HDFS

To copy use the command -cp

Syntax: hdfsdfs -cp source path destination path

```
[cloudera@quickstart ~]$ hdfs dfs -cp /user/cloudera/mydir/file1.txt /user/cloudera/mydir/friends.txt
```

List the files in the destination path

```
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/mydir
Found 3 items
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 22:59 /user/cloudera/mydir/file1.txt
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 23:03 /user/cloudera/mydir/friends.txt
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 23:04 /user/cloudera/mydir/weather.txt
```

7. Rename files in HDFS

```
[cloudera@quickstart ~]$ hdfs dfs -mv /user/cloudera/mydir/weather.txt /user/cloudera/weather_renamed.txt
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/mydir
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 22:59 /user/cloudera/mydir/file1.txt
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 23:03 /user/cloudera/mydir/friends.txt
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/
Found 2 items
drwxr-xr-x   - cloudera cloudera          0 2026-01-19 23:06 /user/cloudera/mydir
-rw-r--r--   1 cloudera cloudera          0 2026-01-19 23:04 /user/cloudera/weather_renamed.txt
[cloudera@quickstart ~]$
```

8. Check disk usage in HDFS

```
[cloudera@quickstart ~]$ hdfs dfs -du -h /user/cloudera/mydir
0  0  /user/cloudera/mydir/file1.txt
0  0  /user/cloudera/mydir/friends.txt
[cloudera@quickstart ~]$ hdfs dfs -du -h /user/cloudera
0  0  /user/cloudera/mydir
0  0  /user/cloudera/weather_renamed.txt
```

Post-Lab:

01. Describe the function of the -p option in the command hdfs dfs -mkdir -p. What occurs if one or more parent directories are missing and this option is not selected?

02. Compare the commands ls / and hdfsdfs -ls /. How does this difference impact data management in a Hadoop cluster, and why do these commands work on different namespaces?

03. The entire HDFS is listed recursively with the command hdfsdfs -ls -R /. During execution, which metadata components are under stress?

04. Describe what occurs internally if the Trash functionality is activated while running hdfsdfs -rm -r /user/cloudera/mydir. What distinguishes this from erasing data in a conventional file system?

05. What is the difference between hdfsdfs -du and hdfsdfs -df? What types of storage information does each command give, and why is this helpful to manage clusters?

(For Evaluator's use only)	
Comment of the Evaluator (if Any)	Evaluator's Observation Marks Secured:_____ out of _____ Signature of the Evaluator